Complex energy learning from finite magnetic-rod observations: coefficient error, local KAM and power guarantees

The International Journal of Robotics Research
XX(X):1–17
©The Author(s) 0000
Reprints and permission:
sagepub.co.uk/journalsPermissions.nav
DOI: 10.1177/ToBeAssigned
www.sagepub.com/
SAGE

Anonymous authors

Abstract

Learning an energy from simulated motion does not by itself establish the derivative accuracy needed for Hamiltonian stability or the power inequalities needed for control. We study a complex-valued network with separate configuration-energy and nonnegative dissipation heads for a finite-element magnetic rod with known constant inertia. A deterministic observation interface converts paired short trajectories at eleven configurations into intervals for 31 quadratic, cubic and quartic energy coefficients under an explicitly declared quadratic material family. A coefficient-guided objective and a precision-weighted final-readout step fit the same trained network. Outward evaluation of the saved parameters verifies all 33 finite-node force and 31 coefficient constraints, with a certified coefficient-loss upper bound below 0.010821. For the same weights and exact source operators, the conservative neural Hamiltonian on all 64 transverse coordinates satisfies a qualitative local elliptic KAM criterion. Residual-controlled modes and a cubic-corrected twist enclosure establish positive-measure families of Lagrangian 64-tori arbitrarily near equilibrium. Saved model settings independently support the quadratic material structure of the ideal benchmark. Exact spectral and twist calculations also establish qualitative local KAM tori for all admitted members of the 64-coordinate transverse conservative family, without an invariant three-coordinate-subspace premise. A nonzero force-loss budget $L_F \leq 12/275$ propagates the selected graph's fitting error to a larger target-parameter interval that also satisfies these conditions. Coherent finite-data ambiguities and a nonzero, cubic-corrected material-jet tolerance quantify the remaining inference gap to an unknown target. Full-coordinate power, coercivity and dissipation are verified separately. A uniform potential-value bound admits all 22 recorded numerical initial states into a fixed energy guard. A sampled-work theorem extends conditional safety to held magnetic inputs through a strict energy ledger; nonzero initial proposals pass at both 1 and $10\,\mu$s for all 22 states. A native three-hold numerical feedback execution is additionally completed. Residual enclosures and the work ledger prove a continuous energy cap for a separately specified full-coordinate reference IVP using its saved initial state, executed fields and switching times. These conclusions require the stated physical family, sensing and actuation conditions. They establish neither unknown weak-nonlinearity identification, perturbative transfer from the learned Hamiltonian to an unknown plant, physical closed-loop implementation nor an advantage over comparable real-valued networks.

Keywords

complex-valued neural networks, Hamiltonian learning, magnetic rods, deterministic error bounds, local KAM theory, dissipative control

1 Introduction

An energy model learned from simulated motion can support a mathematical guarantee only when the observations, learned function and downstream dynamics are defined consistently. Magnetic actuation makes this issue relevant to deformable robots: an applied field supplies mechanical work through a configuration-dependent port, while elasticity and dissipation govern the resulting motion. Magnetic manipulation and ferromagnetic soft continuum robots provide established settings for such interactions (Abbott et al. 2020; Kim et al. 2019). The question studied here is how finite observations can constrain a learned energy sufficiently to support explicitly stated conservative and dissipative conclusions.

Hamiltonian learning supplies a useful structural starting point. A conservative vector field obtained by differentiating

a scalar preserves that scalar along its exact flow (Greydanus et al. 2019). This identity does not establish that the scalar approximates a specified mechanical energy. Finite trajectory observations also leave unresolved dependence away from the observed configurations, and projection of a spatial state does not generally produce an autonomous reduced system. A learning guarantee must therefore retain observation error, the chosen mechanical chart and the distinction between a full-field model and its conservative restriction. The difficulty is especially acute when the nonlinear coefficients of interest are smaller than the uncertainty propagated from the observations.

We study a configuration potential with known kinetic energy. The mechanical object has 64 free transverse coordinates, a constant mass matrix containing translational and rotary inertia, a prescribed magnetic port, and a nominal dissipative matrix. A fixed three-dimensional trial space defines a separate conservative Hamiltonian restriction. The learned full-field correction depends on three projected coordinates and the full nominal elastic energy; on the trial space these four features depend on only three independent configuration variables. Two complex networks represent the potential correction and a nonnegative dissipation factor, with 918 real trainable parameters in total. The kinetic energy remains fixed, so the learned correction has no momentum dependence.

The observation interface retains the full spatial records. Eleven prepared configurations each have two signed-velocity records, giving 22 selected records. Their time samples, configurations, velocities and prescribed fields produce conservative-force labels and signed weak residuals. Continuous enclosures receive each force label against the declared equations initialized by its own recorded numerical state. The bounds retain reconstruction defects, preparation error and arithmetic error. They concern a specified semidiscrete mechanical system; they do not identify a numerical reconstruction with a compiled simulator equation or turn simulated observations into physical measurements.

An exact finite design then connects the 33 force components to the 31 ordinary potential coefficients of degrees two through four. Its error bound propagates componentwise force allowances through an exact left inverse and retains an independent fifth-derivative remainder for a general target. The instantiated calculation uses an explicitly declared quadratic material-correction family, for which this remainder vanishes. This family assumption is separate from the observed labels. The coefficient objective uses the transformed label centers and their received scales, allowing its actual nonzero loss to enter a deterministic candidate-to-target coefficient bound. This finite-data route addresses the nonzero-error question raised by Chen et al. (2022) without assigning an independent sampling law to correlated trajectory data.

The saved candidate is obtained through full-layer training followed by precision-weighted calibration of its final energy readout. With the trained hidden layers fixed, the energy skip and the final complex readout form a 25-dimensional linear fitting block. One numerical least-squares solve calibrates this block within the unchanged architecture. Independent outward evaluation of the saved graph verifies all 33 force and all 31 coefficient interval constraints and supplies a strict upper bound 1.082009×10^{-2} for the nonzero coefficient loss. A separate source-bound calculation verifies the unchanged neural energy on all 64 transverse coordinates: a complete positive spectrum, fourth-order finite nonresonance, residual-controlled modal projections, a nonsingular cubic-corrected twist and an analytic neighborhood imply local Lagrangian 64-tori. The earlier three-coordinate restriction remains a distinct result. Under the explicit hypothesis $\mathcal{W}_* = \beta E_0$, a direct interval calculation also verifies the local criterion for every target Hamiltonian in the admitted scalar interval. A further force-loss interface uses the same saved complex graph: its nonzero weighted force error implies a source-prior target-parameter interval on which all 64 transverse conservative KAM conditions remain valid. This deterministic certificate is more conservative than direct data intersection and makes no population-risk or generic constitutive identification claim. It uses the family prior rather than a perturbative comparison with the learned candidate. For the full-field system, algebraic power identities and a uniform potential-value allowance give a conditional continuous-feedback guard. All 22 saved numerical initial states satisfy that fixed guard. A further work bound for held magnetic inputs yields a conditional sampled-feedback theorem with a strict energy ledger; all 22 nonzero initial proposals are accepted at both 1 and 10 μs hold durations. A further native execution applies two nonzero feedback holds and a scheduled zero hold. A continuous reference-IVP certificate combines residual-enclosed port errors for the two nonzero holds with the dissipative zero-input branch, while preserving the fixed prospective error allowance and recorded commands.

The contribution is a connection between finite observations, an actual nonzero learning objective and conclusions for one specified saved model, with their different hypotheses kept explicit. Its limits are also informative. Exact interval intersection shows that the observations admit a scalar stiffness correction, while their quartic uncertainty exceeds

the weak prescribed magnetic quartic. The data therefore do not require a nonlinear correction or establish an advantage of complex networks. The candidate certificate covers all 64 transverse coordinates of the conservative neural Hamiltonian defined by the saved weights and exact source operators. It requires no invariant trial subspace. Runtime floating-point buffers approximate these operators; they are not identified with exact machine trajectories. A separate calculation establishes local KAM conditions for the declared quadratic target family. Neither result identifies an unknown physical constitutive law. The continuous and sampled safety theorems assume the stated physical family and full-state port. The sampled result also requires valid observations at the actual hold starts, bounded physical damping, and execution of the accepted total field with the prescribed timing. The executed simulation and its reference-IVP certificate supply no physical sensor guarantee or equivalence theorem for the compiled simulator equations. Identification of unknown weak nonlinearities, perturbative learned-to-plant transfer and physical feedback implementation remain separate requirements.

2 Related work

Hamiltonian and port-Hamiltonian learning. Greydanus et al. (2019) parameterize a scalar Hamiltonian and differentiate it to construct a conservative vector field. This separates conservation of the learned scalar from unconstrained vector-field regression. Zhong et al. (2020) extend structured learning to dissipation and external inputs through a port-Hamiltonian computation graph, using observed trajectories and differentiable integration. Desai et al. (2021) likewise use port-Hamiltonian structure to learn systems with time-dependent forcing and dissipation. These works motivate retaining energy, loss and supply as distinct components of a model. We fix the native kinetic and input operators, learn a configuration correction and a scalar dissipation factor, and verify the resulting saved functions independently of their optimizer. The exact power identity is an architectural property; accuracy relative to the specified target enters through separate observation and potential-error bounds.

Complex network representations. Trabelsi et al. (2018) develop practical components for deep complex networks, including complex convolutions, normalization and initialization, and evaluate them on vision and audio tasks. Their results motivate complex affine representations but do not supply a mechanical identification theorem for the present network. Zhang et al. (2022) establish approximation and optimization results for specified complex-reaction networks, including a representation separation for a class of radial functions. Those statements depend on their network and activation assumptions. Our split-tanh implementation is interpreted as a real differentiable graph before continuation in complex configuration variables. Its analytic domain and low-order coefficients are checked for the actual saved parameters. We use neither of these representation references as a generalization bound or a proof of superiority for this architecture. The scalar comparison admitted by the present observations makes that distinction consequential.

Nonzero learning error and conservative invariant tori. Chen et al. (2022) connect nonzero Hamiltonian learning error to KAM theory through gradient-class complexity, statistical generalization and an energy sup-norm estimate. Applying such a route requires its sampling, class and regularity hypotheses. Here the time samples, signed pairs and derived coefficients reuse a fixed finite observation design; they are not independent configuration samples. We instead propagate explicit force allowances to a finite coefficient error and then evaluate the candidate's conservative conditions separately. Classical KAM theory requires arithmetic and nondegeneracy hypotheses in addition to a perturbation estimate (Pöschel 2001). A posteriori formulations can quantify existence near an approximate torus (Haro and Luque 2019). The saved weights and exact source operators define a full 64-coordinate neural candidate treated with a local elliptic criterion using the cubic-corrected quartic normal form. The proof bounds modal residuals before enclosing the actual twist, and retains the generic fifth-order remainder. A separate source-conditioned calculation covers all 64 transverse coordinates of the quadratic target family. The actual nonzero force loss also supplies an outer parameter set on which this direct family criterion is checked. These are qualitative local results, with no specified finite torus domain; the target-family conclusion uses its constitutive prior rather than perturbative transfer of the candidate tori to an unknown material.

Magnetic robotics and the scope of mechanical verification. The survey of Abbott et al. (2020) describes magnetic forces, torques and field generation for remote robotic manipulation. Kim et al. (2019) demonstrate remotely steerable ferromagnetic soft continuum robots. These works provide the robotic context for a mechanical input port whose work must be accounted for alongside elastic storage and

dissipation. The present study addresses a declared semidiscrete beam model and simulated observations. Its full-field power and conditional safety arguments retain the same mass and magnetic port. The full-coordinate conservative neural theorem and its three-coordinate restriction answer separate dynamical questions. A verified mathematical candidate, a finite-element observation record and a physical robot remain distinct objects unless their transfer errors and implementation conditions are separately bounded.

3 Mechanical object and complex energy representation

3.1 Full-coordinate finite-element mechanics

The mechanical object is the transverse finite-element model of a clamped magnetic rod. Its 32 free nodes supply $x = (w_1, \theta_1, \ldots, w_{32}, \theta_{32})^T \in \mathbb{R}^{64}$, with transverse displacement and cross-section rotation at each node. The mesh has spacing 0.0025 m over a length of 0.08 m. The assembled matrices define a particular finite model; no equivalence to an arbitrary geometrically nonlinear continuum rod is assumed. The nominal elastic energy and the known inertia are

$$\begin{aligned} E_0(x) &= \tfrac{1}{2} x^T K_0 x, \\ M = M_0 + M_r \succ 0, &\qquad D_0 = \alpha M_0 \succ 0. \end{aligned} \tag{1}$$

Here M_r is the rotary-inertia contribution, $K_0 \succ 0$ and $\alpha > 0$. The damping matrix uses M_0, rather than the total mass. All quantities in this section use physical SI time and energy. The exact saved finite-element entries define the reference operators; the sum $M_0 + M_r$ is formed before any further rounding. The exact source formulas below define the mathematical graph used in the certificates. Training evaluates once-rounded binary64 operator buffers; the saved network weights and feature scales are interpreted as exact binary constants in the certified graph. No identity with every rounded runtime operation is assumed.

Let $b = e_{64}$ select the tip rotation. For a magnetic field u in tesla and fixed positive coefficients κ and μ, define

$$\begin{aligned} U_0(x) &= E_0(x) + \kappa[1 - \cos(b^T x)], \\ g(x) &= \mu \cos(b^T x) b. \end{aligned} \tag{2}$$

The learned full-state equations are

$$\dot{x} = M^{-1} p = v, \qquad \dot{p} = -\nabla U_N(x) - D_N(x) v + g(x) u. \tag{3}$$

The simulation target used for coefficient evaluation has correction $\mathcal{W}_*(x) = \beta E_0(x)$ and constant damping proportional to D_0. The saved configurations independently specify linear elasticity with geometric nonlinearity disabled. They justify this quadratic material structure for the ideal benchmark, separately from the observed force labels (Supplementary Section S1). The generator uses $E/E_{\mathrm{nom}} = 53/50$; this ratio is not inserted as a training label. The matrices remain the stated exact finite-source constants, without a compiled-runtime equivalence claim. The same constitutive law is not inferred for a general unknown physical device.

3.2 Restriction and its closure boundary

Fix a full-column-rank preparation matrix $V \in \mathbb{R}^{64\times 3}$ and define

$$\begin{aligned} A &= V^T M V, \qquad B = A^{-1} V^T M, \\ K_3 &= V^T K_0 V, \qquad d = V^T b. \end{aligned} \tag{4}$$

Thus $BV = I_3$ and $q = Bx$ is the observed configuration feature. Restricting $x = Vq$ and pulling back the canonical form gives the three-degree-of-freedom Hamiltonian

$$H_N^r(q, p_r) = \tfrac{1}{2} p_r^T A^{-1} p_r + U_N(Vq). \tag{5}$$

The associated full momentum is $p = MVA^{-1} p_r$. This is a Hamiltonian restriction with a well-defined canonical structure. It is not a proof that $\mathrm{range}(V)$ is invariant under the full 64-coordinate dynamics. Full-field observations retain their full coordinates in the label construction and in the power analysis; they are not replaced by a projected autonomous trajectory.

3.3 The two complex-valued heads

Both heads are fully connected $4 \to 12 \to 12 \to 1$ complex networks. For a hidden complex value z, the split activation is $\sigma(z) = \tanh(\mathrm{Re}\, z) + i \tanh(\mathrm{Im}\, z)$. Affine weights and biases are complex; the physical output energy is real. This implementation follows complex affine components and real/imaginary processing as in Trabelsi et al. (2018). Its realification, rather than a claim that a real-valued energy is holomorphic in the original complex hidden variable, defines the analytic continuation used below.

With saved positive scales Q_j and E_{net}, the four real features are

$$s(x) = \big(q_1/Q_1, q_2/Q_2, q_3/Q_3, E_0(x)/E_{\mathrm{net}}\big), \qquad q = Bx. \tag{6}$$

Let $N(s)$ be the real energy readout and $(R(s), I(s))$ the real and imaginary dissipation readouts. Set $\gamma_j = \partial_j N(0)$

for $j = 1, 2, 3$ and define

$$F_N(x) = c_E E_0(x) + E_{\text{net}} \mathcal{N}_c(s(x)), \tag{7}$$

$$\mathcal{N}_c(s) = N(s) - N(0) - \sum_{j=1}^{3} \gamma_j s_j,$$

$$a_N(x) = [1 + c_C + R(s(x)) - R(0)]^2 + [I(s(x)) - I(0)]^2, \tag{8}$$

$$U_N(x) = U_0(x) + F_N(x), \qquad D_N(x) = a_N(x) D_0. \tag{9}$$

The two branches and the real scalars c_E, c_C contain 918 real parameters. Equation (7) enforces $F_N(0) = 0$ and $\nabla F_N(0) = 0$. Only the first three feature derivatives are subtracted. The linear response through the energy feature is retained and contributes to the physical quadratic term. Equation (8) enforces $D_N \succeq 0$ for every parameter vector but does not alone guarantee a positive lower bound.

The configuration feature and the full elastic-energy feature must both be differentiated. Writing $F(q, E)$ for the right-hand side of (7) gives

$$\begin{aligned} \nabla_x F_N &= B^T F_q + F_E K_0 x, \\ V^T \nabla_x F_N &= F_q + F_E V^T K_0 x. \end{aligned} \tag{10}$$

On $x = Vq$, $E_0(Vq) = q^T K_3 q/2$; on a general full trajectory the elastic-energy feature is not determined by Bx alone. This is why the same full observation enters the dissipative weak residual.

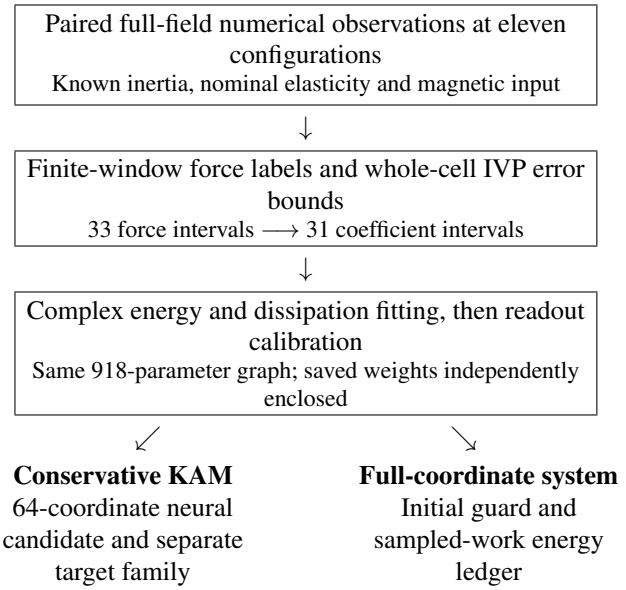


Figure 1. The observation-to-proof chain. With forcing and dissipation removed, the saved weights and exact source operators define a 64-coordinate neural candidate with a source-bound local KAM certificate. Its three-coordinate restriction is verified separately. Under the source-conditioned prior $\mathcal{W}_* = \beta E_0$, a separate proof covers the entire 64-coordinate transverse target family, including the larger parameter set obtained from the actual nonzero force loss. Full-coordinate safety uses a separate potential-value guard and continuous or sampled feedback under explicit physical and execution conditions. All 22 saved numerical initial states satisfy the guard; a further numerical feedback execution receives a continuous cap for a separately specified reference IVP over three holds. Neither branch identifies unknown physics, certifies a physical sensor or establishes complex-network superiority.

4 Finite observations and coefficient error bounds

4.1 Paired trajectories and finite-window force labels

The data are native COMSOL outputs at eleven prescribed configurations. Each configuration has two signed initial velocities and 257 saved full-field observations over $20\,\mu$s. The eight added configurations use $q = \ell z^{(i)}$, $\dot{q} = \pm 25 q\,\mathrm{s}^{-1}$ and internal step 1.953125×10^{-8} s. The preceding three retain their own prescribed velocities and finest saved steps. These choices are documented before their respective computations. The source model is unchanged, and the drive is the source-decimal function

$$B_y(t) = 10^{-4} \cos(11.41692154656479\, t)\ \mathrm{T}. \tag{11}$$

The generator's material is public in the simulation protocol. No material factor or analytic target coefficient is supplied to the learning worker, but this separation is not a hidden-target test.

The finite momentum-and-magnetic-force label approximates a conservative force at a prepared configuration. A trajectory average cannot be identified with its initial value without a temporal allowance. For signed records, let $Y(T)$ be the average obtained from projected momentum increments and magnetic-force quadrature, with the known nominal elastic force subtracted. We use the fixed three-window combination

$$\widehat{f} = \tfrac{8}{3} Y(T) - 2Y(2T) + \tfrac{1}{3} Y(4T). \tag{12}$$

For an exact smooth signal, the constant term is preserved and the first two temporal moments cancel. If its third derivative is bounded by B_3, the remaining temporal bias is at most $B_3 T^3/3$. Preparation defects and finite observation errors must also be included; they are not estimated from a small training residual. The full statement and its nonnegative Peano kernel are proved in Supplementary Section S4.1.

Continuous error is received against an explicitly declared 64-coordinate initial-value problem starting at each saved numerical first row. Its shifted elastic matrix is $K = (53/50)K_0$, and its damping is $D = \alpha_{\mathrm{shift}} M_0$, with the saved binary coefficient $\alpha_{\mathrm{shift}} \simeq 0.6389230223455499\,\mathrm{s}^{-1}$. These are the declared evaluation-IVP coefficients, not extra material labels given to the learner. A cubic Hermite curve through saved positions and velocities has a computable whole-cell equation defect. In the mechanical energy metric the trajectory-error radius obeys

$$r'(t) \le a r(t) + \rho(t), \qquad r^2 = e_x^T K e_x + e_v^T M e_v, \quad (13)$$

where ρ bounds the mass-dual equation defect. Outward Bernstein-polynomial bounds, trigonometric remainders and the exact source-decimal input enclose each whole time cell. A projected-force average retains cancellations that a generic endpoint-momentum bound would discard. The resulting received label allowances are below 4.583×10^{-8} in normalized force at all eleven chosen configurations. This certifies the declared recorded-state IVPs; it does not certify physical measurement error or equivalence to every compiled COMSOL operation. Supplementary Sections S4.3 and S4.8 give the full construction.

4.2 The eleven-node fourth-order design

Set $q = \ell z$, $\ell = 10^{-3}$ m, and use the exact decimal energy scale $E_s = 6 \times 10^{-12}$ J for the force/coefficient interface. The saved binary scale E_{net} is kept distinct from E_s. For the restricted correction $W_*(q) = \mathcal{W}_*(Vq)$, define $w_*(z) = W_*(\ell z)/E_s$. The normalized force labels approximate $\nabla_z w_*(z^{(i)})$ at the ideal rational nodes, while the learner's once-rounded input coordinates are recorded separately.

Under $\nabla w_*(0) = 0$ and a componentwise fifth-derivative bound M_5, its ordinary Taylor coefficients of degrees two through four form $c_* \in \mathbb{R}^{31}$. With the fixed gradient design matrix $G \in \mathbb{Q}^{33\times 31}$ and exact left inverse J, one has

$$\begin{aligned} JG &= I_{31}, \\ |c_{*,\alpha} - (J\widehat{y})_\alpha| &\le \sum_{i,j} |J_{\alpha,(i,j)}| \varepsilon_{ij}, \\ \varepsilon_{ij} &= \eta_{ij} + \frac{M_5}{24} \|z^{(i)}\|_1^4. \end{aligned} \quad (14)$$

Here $\widehat{y} \in \mathbb{R}^{33}$ contains the saved force midpoints and η includes each component's continuous and arithmetic allowance. The exact full-rank construction and tail derivation are in Supplementary Proposition S11. The fifth-derivative bound is an independent target hypothesis. For the explicitly declared family $W_* = \beta q^T K_3 q/2$, $M_5 = 0$ and the origin gradient vanishes by that constitutive assumption.

Writing $\widehat{c} = J\widehat{y}$, the largest received radii for ordinary coefficients of degrees two, three and four are

$$\begin{aligned} \max_{|\alpha|=2} r_\alpha &\le 4.282817 \times 10^{-7}, \\ \max_{|\alpha|=3} r_\alpha &\le 8.141379 \times 10^{-7}, \\ \max_{|\alpha|=4} r_\alpha &\le 1.808799 \times 10^{-6}, \end{aligned} \quad (15)$$

respectively, rounded upward. These are deterministic error intervals from the same 33 observations, not 31 additional independent samples. A small force error does not by itself bound arbitrary higher derivatives; the design, centering and remainder hypotheses in (14) are essential.

4.3 Nonzero-loss bounds for the saved network

Let $c_{\theta,\alpha}$ be the ordinary normalized correction coefficients of the actual saved graph. The mathematical data centers are $\widehat{c} = J\widehat{y}$. Their saved training centers $\widetilde{c}$ are binary numbers. Exported radii satisfy $\widetilde{r}_\alpha \ge r_\alpha + |\widetilde{c}_\alpha - \widehat{c}_\alpha|$, including the exact center-rounding allowance. With the fixed saved scales $\widetilde{s}_\alpha = \max(\widetilde{r}_\alpha, 10^{-8})$, the actual coefficient objective is

$$\widetilde{L}_{\mathrm{jet}}(\theta) = \frac{1}{31} \sum_\alpha \left(\frac{c_{\theta,\alpha} - \widetilde{c}_\alpha}{\widetilde{s}_\alpha} \right)^2. \quad (16)$$

The data interval gives, for every parameter vector,

$$|c_{\theta,\alpha} - c_{*,\alpha}| \le |c_{\theta,\alpha} - \widetilde{c}_\alpha| + \widetilde{r}_\alpha \le \widetilde{s}_\alpha \sqrt{31 \widetilde{L}_{\mathrm{jet}}(\theta)} + \widetilde{r}_\alpha. \quad (17)$$

No optimizer convergence or independent-sampling assumption is used. Each component's direct residual normally gives the tighter first bound. The exact original-center version is also valid with $\widehat{c}$, r and any fixed positive scales, but is not silently substituted for the saved-center objective actually optimized.

Float64 losses guide optimization but are not proof bounds. For an outward coefficient enclosure $[a_\alpha, b_\alpha]$, define $\Delta_\alpha = \max(|a_\alpha - \widehat{c}_\alpha|, |b_\alpha - \widehat{c}_\alpha|)$. Then $|c_{\theta,\alpha} - c_{*,\alpha}| \le \Delta_\alpha + r_\alpha$. Replacing $\widehat{c}_\alpha$ by $\widetilde{c}_\alpha$ in the endpoint residual gives $\widetilde{\Delta}_\alpha$; then $(1/31)\sum(\widetilde{\Delta}_\alpha/\widetilde{s}_\alpha)^2$ is a rigorous upper bound for the actual saved-center loss. Supplementary Section S8 proves this finite-loss interface and distinguishes original from rounded training centers.

The configuration-potential formulation also permits a conditional population-risk route inspired by Chen et al. (2022). That route requires independent configuration

samples, a justified bounded loss class, a density lower bound and target regularity. The eleven deterministic preparations do not instantiate those statistical assumptions. The present numerical conclusions use (14)–(17); the separate population estimates and their prerequisites are retained in Supplementary Section S2.

5 Training and precision-weighted coefficient fitting

5.1 Supervision of energy and dissipation

The energy head is supervised by the eleven observed conservative force labels and their derived coefficient intervals. Its point prediction is the normalized gradient of $F_N(Vq)$, including the elastic-energy feature derivative in (10). The dissipation head uses the odd part of paired weak momentum residuals on the 22 full-coordinate records. Each signed record supplies four 64-panel windows. An individual weak residual includes momentum endpoints and an integral of conservative, dissipative and magnetic forces. The signed difference emphasizes the velocity-dependent term but retains the actual imperfect initial symmetry and numerical observations. It is not assumed to be an exact isolated damping measurement.

The conservative contribution is detached during dissipation updates. Thus energy and dissipation are fitted by specified block objectives; the whole procedure is not described as complete gradient descent on a single coupled selection score. All eleven configurations are development data. Neither their 257 temporal samples nor their velocity signs are counted as independent configuration samples.

5.2 Full-layer fitting and its high-order failure

The network is initialized from a preceding single-configuration fit that performed 200 Adam updates and selected step 191. Eleven-configuration full-batch training then performs 2000 Adam updates at learning rate 10^{-3} and seed 43, selecting step 2000 by the combined same-data score. Both real and imaginary parts of all three weight layers in both branches change. The maximum normalized point-force residual is approximately 1.563110×10^{-5}. Direct outward evaluation nevertheless finds a quartic coefficient discrepancy larger than 6.345877×10^{-4} against the declared quadratic material family. This comparison is made after training; analytic zero cubic or quartic values are not substituted as labels.

Adding (16) reduces that high-order discrepancy. The energy branch uses one strong-Wolfe LBFGS iteration per outer cycle, and the dissipation branch one Adam step at fixed current energy. All 200 prescribed outer cycles are completed, with 432 energy closure evaluations and 200 dissipation updates. The coefficient loss falls from about 50743.76 to 4.399638. However, the point-force residual rises to approximately 2.434500×10^{-4}. In this stage the point residual is divided by the observed signal RMS whereas the coefficient residual is divided by its much smaller received error. A small coefficient objective therefore does not ensure an adequate finite-node fit under that scaling.

5.3 A linear block in the unchanged complex graph

Fixing the two trained hidden layers makes the energy correction linear in c_E and the 24 real coordinates of its final complex weight. The centered output bias is a gauge and remains fixed. Let $\xi \in \mathbb{R}^{25}$ contain these coordinates and $A_\psi \xi \in \mathbb{R}^{64}$ contain the 33 force and 31 coefficient predictions. The final calibration solves the numerical least-squares problem

$$\min_{\xi} \left\| D^{-1} \left(A_\psi \xi - \begin{pmatrix} \widehat{y} \\ \widetilde{c} \end{pmatrix} \right) \right\|_2^2, \qquad D = \operatorname{diag}(\eta, \widetilde{s}). \tag{18}$$

Here $\widetilde{c}$ and $\widetilde{s}$ are the saved coefficient centers and scales including center-rounding allowances. Each term is $(\text{residual}/\text{scale})^2$. There is no ridge penalty, and the force scales are their own received observation errors. The numerical design has rank 25 and condition estimate 2.84×10^6. These are floating-point diagnostics, not an exact-rank or exact-optimality proof. The exact-real fit is verified after saving the selected weights.

This is one final-readout solve and zero gradient updates. Only c_E and the final energy weight change; all other coordinates, including the entire dissipation head, remain identical to the preceding trained state. The model retains its 918-parameter graph. The method and scaling were chosen after observing the preceding fit and are reported as adaptive development, without a blind validation claim. The 64 rows in (18) do not represent 64 independent observations. Supplementary Section S8.1 proves the linear-block identity for the same centered network.

5.4 Independent reception of the selected parameters

Arb evaluation at 512 bits encloses the exact-real outputs defined by the saved binary weights, saved scales and

Table 1. Upward bounds on normalized ordinary correction-coefficient error. The first numerical column uses the data intervals and the triangle inequality. The second additionally evaluates the declared quadratic simulation target.

Degree	Data-based bound	Declared-family evaluation
2	4.616863×10^{-7}	3.635946×10^{-8}
3	8.174276×10^{-7}	1.768735×10^{-8}
4	2.066906×10^{-6}	3.908709×10^{-8}

exact native operators. At all eleven ideal rational nodes the 33 force enclosures satisfy their respective received intervals. All 31 coefficient enclosures also satisfy the original data intervals. The maximum normalized point-force residual is below 1.797797×10^{-8} and its largest ratio to its received allowance is below 0.790301. Including the admitted observation error gives the finite-node candidate-to-target force bound 5.023184×10^{-8}. This bound is restricted to the eleven configurations and to the declared observation/target assumptions.

The independently computed loss upper bound is $\overline{L}_{\mathrm{jet}} < 0.01082009$. Table 1 distinguishes data-based guarantees from the tighter evaluation available only for the declared material formula. In particular, the smaller evaluator error must not be presented as the information content of the observed coefficient intervals.

The final energy skip is about 0.06002256805 and the norm of the 24 real readout coordinates is about 0.0017053353. The raw skip is not itself an identified material factor, because the neural energy-feature response also contributes to the quadratic term. An exact rational feasibility calculation admits the scalar comparison family $W(q) = \zeta q^T K_3 q/2$ with nonempty compatible interval approximately $[0.0599999981693, 0.0600000004402]$. An exact rational midpoint satisfies all 33 force and 31 coefficient constraints without assigning the material value to ζ. Thus the finite data do not require a nonlinear neural correction. Detailed source-bound coefficient and feasibility calculations appear in Supplementary Section S9.1.

6 Conservative learning error and local KAM certification

6.1 What the coefficient bound can certify

For the restriction (5), known kinetic and reference magnetic terms are shared between the learned and target Hamiltonians. Their vector-field difference therefore depends only on the conservative correction force. The learning variable can be three-dimensional configuration, even though the canonical phase space has six real dimensions. This does not reduce the mechanical number of degrees of freedom. Under independent configuration sampling and suitable class and regularity conditions, the configuration-risk estimates in Supplementary Section S2 provide one connection with the nonzero-loss theory of Chen et al. (2022).

The deterministic bound (17) supplies a different, finite-data interface. Multiplication by $E_s \ell^{-|\alpha|}$ gives physical ordinary monomial coefficients; the derivative $D^\alpha W(0)$ additionally includes the factor $\alpha!$. The resulting Hessian, cubic and quartic intervals could support a physical-target KAM conclusion only if they imply the required spectral, nonresonance and twist conditions uniformly, together with the target's analytic hypotheses. Without additional constitutive structure, the present quartic intervals do not establish that implication. Their radius is much larger than the largest prescribed normalized magnetic quartic coefficient, approximately 5.363314×10^{-9}. All fifteen admitted total quartic coefficient intervals contain zero. This sign-ambiguity diagnostic is not itself a calculation of the Birkhoff twist.

6.2 A criterion for the three-coordinate restriction

The selected network can also be verified as a Hamiltonian in its own right. Suppose $U_N(Vq)$ has a realified holomorphic continuation near the origin, its Hessian is positive definite, and its positive linear frequencies obey $k \cdot \omega \neq 0$ for integer vectors $0 < |k|_1 \leq 4$. The cubic terms are removed by the homological equation. With cubic generating function χ_3, the effective quartic term is

$$H_{4,\mathrm{eff}} = H_4 + \tfrac{1}{2}\{H_3, \chi_3\}. \tag{19}$$

Its action projection has the form

$$h_N(I) = \omega \cdot I + \tfrac{1}{2} I^T T_N I, \qquad \det T_N \neq 0. \tag{20}$$

Under these hypotheses, the classical local elliptic KAM argument gives a positive-measure family of invariant Lagrangian three-tori in every sufficiently small neighborhood of the equilibrium. Supplementary Proposition S17 gives the normal-form calculation and the scaling needed to apply Pöschel (2001). The finite frequency search is not an all-integer Diophantine assertion about the equilibrium frequencies: the Diophantine frequencies of the resulting tori vary in the nonlinear frequency image.

6.3 Same-checkpoint restriction result and its scope

The selected graph is pole-free on the explicit complex polydisc $|z_j| \le 1/64$, with $q = \ell z$. Its positive mass and Hessian, finite degree-three/four divisors and cubic-corrected twist are received afresh from the same saved parameters. The frequencies are approximately

$$(11.73745102,\ 72.26615415,\ 326.09695871)\ \mathrm{rad\,s^{-1}}. \tag{21}$$

The twist has two positive and one negative eigenvalues and determinant near -5.271108×10^{19}, excluding zero. These checks meet the preceding qualitative criterion for this same conservative restriction. They do not specify a nonzero finite torus radius or certify a trajectory as an individual torus. The radius $1/64$ is an analytic-domain bound, not a torus radius.

This result concerns the learned conservative restriction with dissipation and time-dependent forcing removed. It establishes neither perturbative transfer to a physical target, invariance of the three-coordinate subspace inside the full rod, nor torus persistence under the dissipative controller. The following direct target-family calculation uses an additional constitutive hypothesis.

6.4 The neural Hamiltonian on all transverse coordinates

The full conservative candidate is defined by the same weights and exact source operators of Section 3:

$$H_N^{64}(x,p) = \tfrac{1}{2} p^T M^{-1} p + U_N(x), \qquad (x,p) \in \mathbb{R}^{128}. \tag{22}$$

This retains the full energy feature $E_0(x)$ and all 64 transverse configuration variables. It requires no invariance of the trial space. Let $P = \mathrm{diag}(1/Q)B$, $E = E_{\mathrm{net}}$, and write the realified energy readout as $N(l,e)$. Its actual Hessian is

$$\begin{aligned} U_N''(0) &= \alpha_N K_0 + E P^T N_{ll}(0) P + \kappa b b^T, \\ \alpha_N &= 1 + c_E + N_e(0). \end{aligned} \tag{23}$$

The energy-feature derivative is retained. Exact mass-metric bounds and spectral isolation prove 64 positive simple frequencies, with

$$|k \cdot \omega_N| \ge \frac{545883796425}{281474976710656} > 0.00193936\ \mathrm{rad\,s^{-1}} \tag{24}$$

for every integer k with $0 < |k|_1 \le 4$.

The normal modes used for the nonlinear calculation are not accepted as exact floating-point eigenvectors. An exact proposal Gram matrix, Rayleigh residuals and separated spectral intervals give rigorous enclosures of the actual modal feature and tip projections. These enclosures propagate all energy-feature chain terms and the cubic homological correction into the full quartic action Hessian T_N^{64}. For $\Omega_N = \mathrm{diag}(\omega_{N,i})$, an outward product with a retained preconditioner B_0 proves

$$\left\| I - B_0 \Omega_N^{-1} T_N^{64} \Omega_N^{-1} \right\|_\infty < 2.353 \times 10^{-6} < 1. \tag{25}$$

Thus T_N^{64} is nonsingular. The realified graph is pole-free on a full complex configuration polydisc with displacement and rotation radii 2^{-40} m and 2^{-40} rad, respectively; momenta are entire. This is an analytic-domain certificate, not a torus-amplitude bound.

Supplementary Theorem S20 combines these compatible inputs. The saved graph has nonzero cubic terms, so its fourth-order normal form has a general fifth-order remainder. The small-action scaling applies classical KAM theory without an evenness assumption. Every open neighborhood of the equilibrium consequently contains a family of analytic invariant Lagrangian 64-tori whose union has positive 128-dimensional measure. The conclusion concerns this fixed conservative source graph. It includes no axial coordinates, damping, forcing, numerical torus radius, unknown-material transfer or superiority claim. Runtime operator rounding remains a separate implementation issue.

6.5 A direct local result for the admitted target family

Assume explicitly $\mathcal{W}_* = \beta E_0$, with β in the entire exact scalar compatibility interval of Section 5; its endpoints are approximately 0.0599999981693 and 0.0600000004402. The conservative three-coordinate target restriction is then

$$H_\beta^r(q,p_r) = \tfrac{1}{2} p_r^T A^{-1} p_r + \tfrac{1+\beta}{2} q^T K_3 q + \kappa[1 - \cos(d^T q)]. \tag{26}$$

Its cubic part vanishes. If u_j are mass-orthonormal linear modes, the quartic twist in physical canonical actions depends on $a_j = (d^T u_j)^2/\omega_j$ and obeys

$$\det T_\beta = -\frac{5\kappa^3}{512}(a_1 a_2 a_3)^2. \tag{27}$$

Uniform interval enclosures verify a positive simple spectrum, strictly positive modal port overlaps, and all frequency divisors through order four over the complete parameter interval. They give $|k \cdot \omega| > 11.73745099\ \mathrm{rad\,s^{-1}}$

for $0 < |k|_1 \le 4$, together with

$$-6.217103 \times 10^{17} < \det T_\beta < -6.217101 \times 10^{17}. \quad (28)$$

The twist has two positive and one negative eigenvalues throughout. Every admitted family member therefore has positive-measure families of invariant Lagrangian three-tori in sufficiently small neighborhoods of its equilibrium. Supplementary Corollary S22 supplies the uniform calculation and its analytic hypotheses.

This is a direct qualitative theorem for the declared family; it does not assert a numerical torus radius or an all-integer Diophantine property of the equilibrium frequencies. It also does not identify that family for an unknown device, prove perturbative transfer from the learned candidate, or establish invariance of the trial subspace within the full 64-coordinate system. The candidate certificate and the target-family certificate consequently answer distinct questions.

6.6 Source-conditioned extension to all transverse coordinates

The ideal benchmark's saved linear-elastic, small-rotation settings provide independent evidence for $\mathcal{W}_* = \beta E_0$ (Supplementary Section S1). For this constitutive family, retain all 64 transverse coordinates and define

$$H_\beta(x,p) = \tfrac{1}{2} p^T M^{-1} p + (1+\beta)E_0(x) + \kappa[1-\cos(b^T x)]. \quad (29)$$

Over the entire original scalar compatibility interval, exact rational inertia calculations isolate all 64 positive linear frequencies. A degree-preserving finite-field coprimality certificate excludes zero modal tip overlaps. The uniform finite-divisor bound is

$$|k\cdot\omega(\beta)| \ge \frac{2472231}{1073741824} > 0.00230244\ \mathrm{rad\,s^{-1}}, \quad (30)$$

for every integer k with $0 < |k|_1 \le 4$. Writing $a_j = (b^T u_j)^2/\omega_j$ for mass-orthonormal modes, the quartic action Hessian is

$$T_\beta = \tfrac{\kappa}{8}\,\mathrm{diag}(a)(I_{64} - \mathbf{2}\mathbf{1}^T)\,\mathrm{diag}(a). \quad (31)$$

All a_j are nonzero. Hence T_β has inertia $(63,1,0)$ and nonzero determinant throughout the interval. Supplementary Proposition S23 gives the exact certificates. The small-action scaling in Supplementary Theorem S24 then applies Pöschel (2001) in dimension 64 and gives positive-measure families of invariant Lagrangian 64-tori arbitrarily near equilibrium.

These are tori in the 128-dimensional transverse phase space. The calculation uses neither invariance of the three-coordinate trial space nor a claim about 96-dimensional Lagrangian tori with axial dynamics. It assigns no numerical torus radius and makes no claim about damped or forced torus persistence, compiled-solver equivalence, or an unknown physical material. The independent full-coordinate neural certificate in Section 6.4 concerns a different Hamiltonian and uses its own cubic-corrected twist.

6.7 From the nonzero force-loss block to target-family KAM

The same prior allows the actual network fit to enter a sufficient target-family certificate. Let $g_{\theta,j}$ be its exact normalized force at observation component j, and use the received allowance η_j in the force block of the precision-weighted objective:

$$L_F(\theta) = \frac{1}{33}\sum_{j=1}^{33}\left|\frac{g_{\theta,j}-\widehat{y}_j}{\eta_j}\right|^2. \quad (32)$$

This is a force-block score on the existing development observations, distinct from the 31-coefficient loss. If $\overline{L}_F$ is the strict upper bound obtained from the saved output enclosures and the target force is βm_j, where m_j is the known observation multiplier for $j = 1,\dots,33$, the triangle inequality gives

$$|g_{\theta,j} - \beta m_j| \le \eta_j\big(1 + \sqrt{33\overline{L}_F}\big). \quad (33)$$

For the same unchanged checkpoint,

$$0.03 < L_F \le \overline{L}_F < 0.037838 < 12/275. \quad (34)$$

Hence $11\eta_j/5$ is a valid allowance in (33). Intersecting its candidate-centered outer parameter strips gives an exact rational interval with outward decimal enclosure $[0.05999999283191808, 0.06000000327884982]$. The original data-only interval is contained in this larger interval; the learning score supplies a sufficient certificate without adding identification information to the observations.

The spectral calculation is repeated over the enlarged interval, giving $|k\cdot\omega| \ge 2646043/2147483648 > 0.00123215$ for $0 < |k|_1 \le 4$. The parameter-uniform certificate of nonzero modal overlaps and the twist formula (31) then give the same qualitative local 64-torus conclusion for every admitted target. Supplementary Propositions S26 and S27 prove the loss interface and its numerical instance.

This connects the actual nonzero force loss to a target-parameter set and then to a full-transverse conservative conclusion under the stated global constitutive prior. This loss-to-family construction does not itself prove the neural candidate's full-coordinate KAM property; the independent calculation in Section 6.4 supplies that result. The bound $12/275$ is a posteriori sufficient for this checkpoint and these observations; it is neither necessary nor a training stopping rule specified in advance. No generic nonlinear identification or complex representation advantage follows.

6.8 Coherent ambiguity and a nonzero material-jet neighborhood

The marginal sign ambiguity can be strengthened to a single coherent alternative. The material correction $\widetilde{W}(q) = \widetilde{\beta}E_0(q) + \kappa(d^T q)^4/24$ admits one common parameter $\widetilde{\beta}$ satisfying all 33 original force-component and 31 derived coefficient intervals simultaneously. Its fifth material derivative is identically zero, while the retained magnetic cosine makes the total cubic and quartic Hamiltonian terms vanish. Its degree-four Birkhoff twist is therefore exactly zero (Supplementary Proposition S13). This coherent witness is admitted after allowing quartic material corrections; it does not contradict the explicitly quadratic family theorem. Independently, an even analytic correction with a nonzero higher-order tail leaves every retained signed-node force exactly unchanged and cancels the same quartic jet (Supplementary Proposition S14). Thus precise finite forces do not replace an independent target-tail condition. Neither construction proves absence of all invariant tori.

A positive sufficient result permits general cubic and quartic material perturbations while fixing the quadratic jet exactly. Let η_3, η_4 bound their ordinary coefficients in the normalization $q = \ell z$, W/E_s. Including the cubic homological correction, the full admitted family retains nonsingular twist whenever

$$\begin{aligned} b_4\eta_4 + b_3\eta_3^2 &< 1, \\ b_4 &< 1.123431 \times 10^{11}, \\ b_3 &< 2.663083 \times 10^{12}. \end{aligned} \tag{35}$$

The rectangle $(\eta_3, \eta_4) = (2^{-22}, 2^{-38})$ gives a bound below 0.561; the common radius $\eta_3 = \eta_4 = 2^{-38}$ gives a bound below 0.409. Supplementary Lemma S28 establishes the criterion, and independent real-coordinate normal-form calculations verify the numerical instance. These are nonzero sufficient coefficient neighborhoods for a qualitative local theorem, with no numerical torus radius assigned.

The observed material coefficient boxes, compared directly with the same quadratic-family reference, give radii approximately $8.168713334 \times 10^{-7}$ and $2.067036334 \times 10^{-6}$ for degrees three and four. Their sufficient inverse-margin majorant exceeds 2.32218×10^5 and does not meet (35). These are material-to-family quantities, distinct from a candidate-to-target error bound. In addition, the finite-data quadratic uncertainty and the unknown material's target-tail and analytic assumptions have not been received by this fixed-quadratic theorem. The current learning-to-KAM transfer to an unknown physical target therefore remains unproved. Failure of this sufficient bound is not a conclusion that the target lacks tori.

7 Full-coordinate power and conditional magnetic safety

7.1 Power identity and positive dissipation

Let $H_N(x, v) = v^T M v/2 + U_N(x)$ and let $y = g(x)^T v = \mu \cos(b^T x) b^T v$ be the full-state magnetic port. Equations (3) and (10) give

$$\dot{H}_N = -a_N(x) v^T D_0 v + uy. \tag{36}$$

This identity uses the exact constant mass and the actual full state, including the energy-feature chain rule. It holds for all network parameters and is independent of the restricted KAM result. It expresses structure preservation, not identification of the physical dissipative force. No learned momentum correction enters $\dot{x} = M^{-1}p$, so the candidate and plant ports agree if their mass, input map and full state agree.

For the selected weights, $c_E \simeq 0.06002256805$ and $c_C \simeq 0.07237264643$. The condition $1 + c_E > 0$, bounded real split-tanh output and a quadratic completion prove real coercivity of the candidate energy. Exact native factorizations give $D_0 > (6/25)M\,\mathrm{s}^{-1}$. On the full-state feature box $|s(x)|_\infty \le 2$, exact saved-weight bounds give

$$a_N(x) > 0.3977230, \qquad D_N(x) > 0.0954535 M\,\mathrm{s}^{-1}. \tag{37}$$

These are same-checkpoint lower bounds. The feature box is not automatically the full configuration energy domain used below. Supplementary Section S16.1 gives the other feature radii and exact conventions.

7.2 A potential-value safety theorem for the physical plant

Consider a physical system with the same mass and magnetic map, potential U_*, locally Lipschitz damping $D_*(x) \succeq 0$, and no additional unmodeled work. Suppose on the closure of a bounded configuration domain Ω,

$$|U_N - U_*| \le \varepsilon_U. \tag{38}$$

For actual continuous feedback $u = -\operatorname{sat}_U(ky)$, $U, k > 0$, assume

$$x(0) \in \Omega, \quad H_N(0) + \varepsilon_U \le c, \quad \inf_{\partial\Omega} U_N - \varepsilon_U > c. \tag{39}$$

Then the physical trajectory exists for all positive time, remains in Ω and obeys

$$H_*(t) \le H_*(0) \le c, \qquad H_N(t) \le c + \varepsilon_U. \tag{40}$$

Indeed, $\dot{H}_* = -v^T D_* v - y \operatorname{sat}_U(ky) \le 0$. The initial value estimate gives $H_*(0) \le c$. A first exit at the configuration boundary would require $H_* \ge U_* > c$, contradicting the energy inequality. Compactness and positive mass bound the velocity and allow continuation. This proof requires a uniform potential-value allowance; a finite-node force fit alone does not supply it. Supplementary Theorem S30 states the precise smoothness and domain hypotheses and gives the complete proof.

The physical and learned storage derivatives need not coincide. Writing $G = \nabla(U_N - U_*)$ gives, along the physical plant, $\dot{H}_N = -v^T D_* v + uy + v^T G$. Consequently a learned-energy sublevel is not automatically invariant even when the physical energy decreases. The conclusion (40) uses the potential-value comparison and the physical storage, rather than assuming away this discrepancy.

7.3 A same-checkpoint conditional guard

For a numerical instance, assume the physical family

$$U_*(x) = (1+\beta)E_0(x) + \kappa[1 - \cos(b^T x)], \tag{41}$$

where the constant β lies in the exact scalar compatibility interval of Section 5. This is an explicit constitutive-family and interval assumption. Simulation-derived compatibility is not evidence that an arbitrary physical rod necessarily belongs to that family.

On $\Omega_4 = \{x : E_0(x) < 4E_{\mathrm{net}}\}$, recomputed global derivative bounds and outward origin evaluation for the selected energy head prove

$$\sup_{\overline{\Omega}_4} |U_N - U_*| < 0.003838998456E_{\mathrm{net}}, \tag{42}$$

$$\inf_{\partial\Omega_4} U_N > 4.236161085119E_{\mathrm{net}}. \tag{43}$$

Set $c = 2.05E_{\mathrm{net}}$. The boundary guard has positive normalized margin 2.182322086663. With $h_0 = 2.046161001544E_{\mathrm{net}}$, the initial set

$$\mathcal{I} = \{(x, v) : x \in \Omega_4, \quad H_N(x, v) \le h_0\} \tag{44}$$

contains the origin and an open neighborhood. Thus (40) follows conditionally for every initial state in $\mathcal{I}$ under the specified continuous full-state feedback. The global energy-head Hessian, compliance and scalar-interval calculation are supplied directly in Supplementary Section S16.2.

All 22 finest signed records at the eleven configurations belong to this same fixed initial set. Each calculation uses its own saved $t = 0$ full-coordinate position and velocity, interpreted as exact binary64 values, with no replacement by Vq or by the opposite sign. The strict common bounds are

$$\begin{aligned} 0.237377387997 &< H_N(x_0, v_0)/E_{\mathrm{net}}, \\ H_N(x_0, v_0)/E_{\mathrm{net}} &< 0.972247611362. \end{aligned} \tag{45}$$

$$\frac{c - \varepsilon_U - H_N(x_0, v_0)}{E_{\mathrm{net}}} > 1.073913390182. \tag{46}$$

Supplementary Corollary S33 gives the separate sign-wise bounds and native coordinate reception. These are numerical initial states, with no certified error bound to a true physical initial state.

7.4 A sampled-work bound and strict energy ledger

For sampled control, assume additionally the physical damping form $D_*(x) = a(x)D_0$, with locally Lipschitz a and $0 \le a(x) \le 6/5$. This upper operator bound is a physical hypothesis separate from (37). Let u_k be the actual total transverse field held for duration h_k, with $|u_k| \le U$. It replaces the preparation cosine drive. On $H_* \le c$, the full-state equation and the energy cap give an output-rate bound $|\dot{y}| \le L_y$. For the present native matrices and $U = 10^{-4}$ T, outward evaluation proves

$$L_y < 49.953396682 \ \mathrm{J\,T^{-1}s^{-2}}. \tag{47}$$

At a nonzero hold's actual start t_k, require a valid current observation $|y(t_k) - \widehat{y}_k| \le \varepsilon_k$. Since $\dot{H}_* = -a(x)v^T D_0 v + u_k y \le u_k y$, every hold prefix $0 \le s \le h_k$ below the energy

cap satisfies

$$H_*(t_k+s)-H_*(t_k)\le F_k(s),\qquad F_k(s):=s(u_k\widehat{y}_k+|u_k|\varepsilon_k)+\tfrac{1}{2}s^2|u_k|L_y. \tag{48}$$

The omitted dissipation is nonnegative. This choice takes zero dissipation credit; it does not assume a zero derivative of the physical damping coefficient.

Initialize a trusted ledger by rounding an upper enclosure of $H_N(x_0,v_0)+\varepsilon_U$ upward to a fixed energy grid g, so $H_*(t_0)\le L_0<c$. Write $\Delta_k=F_k(h_k)$. Accept a nonzero proposal only if its observation and actual field limit are valid and

$$\Delta_k\le\alpha_{\mathrm{res}}(c-L_k),\qquad 0<\alpha_{\mathrm{res}}<1, \tag{49}$$

$$L_{k+1}=\lceil\max\{0,L_k+\Delta_k\}\rceil_g<c. \tag{50}$$

Otherwise apply exactly zero and retain the ledger. An exactly zero total-field hold also uses this dissipative branch directly and requires no port-observation error bound. Starting with $H_*(t_k)\le L_k<c$, convexity of F_k yields $F_k(s)\le\max\{0,\Delta_k\}$. A hypothetical first time with $H_*=c$ would therefore satisfy $H_*\le L_k+\max\{0,\Delta_k\}<c$, a contradiction; the rate bound is used only before that first exit. At the hold endpoint, $H_*(t_{k+1})\le L_{k+1}<c$. Induction gives the strict energy cap for any sequence of holds without finite-time accumulation, provided the observation, timing and total-field execution conditions hold at every step. Also $c/(1+\beta_-)<4E_{\mathrm{net}}$, so this physical cap keeps the configuration in Ω_4. Supplementary Section S17 gives the rate majorant, continuation argument and complete execution assumptions.

For a received initial-proposal calculation, fix $\varepsilon_y=10^{-8}\,\mathrm{J\,T^{-1}s^{-1}}$, $g=10^{-9}E_{\mathrm{net}}$ and $\alpha_{\mathrm{res}}=1/2$. At each of the 22 saved numerical states, an outward port enclosure supplies a rational midpoint $\widehat{y}_0$ whose arithmetic error is below this allowance. The nonzero opposite-sign proposal $u_0=-U\,\mathrm{sign}(\widehat{y}_0)$ passes (49)–(50) for all 22 states at both $1\,\mu$s and $10\,\mu$s. At $100\,\mu$s, all 22 proposals are rejected and the exact zero fallback retains their initial ledgers. The same cap, field limit, error allowance and rounding grid are used throughout; Supplementary Section S17.1 gives the corresponding ledger and work bounds.

The port-error allowance is conditional for a physical sensor, and this initial-proposal calculation alone executes no feedback rollout; Section 7.5 gives a separate numerical execution. Accepted holds may spend energy reserve; neither monotone energy decrease nor improved control performance is asserted. The arbitrary-horizon theorem depends on valid sensing at each actual hold start and execution of each accepted total field with the specified timing. Physical sensing, actuator and timing guarantees remain to be supplied for an experiment.

7.5 Executed feedback and a continuous reference-IVP certificate

A new native COMSOL execution uses the selected network's potential guard to initialize the energy ledger and the common mechanical port to choose the field sign. Starting from a fresh $20\,\mu$s preparation solve, two approximately $10\,\mu$s feedback holds apply opposite nonzero total fields, followed by a scheduled zero-field hold. The next sign is computed from the current saved state; the preparation cosine is replaced during each hold. The executed magnitudes are the binary64 number immediately below 10^{-4} T. All 101 native solution and velocity entries, including the clock variables, agree bitwise at the three continuation handoffs. This is a numerical feedback execution, with no physical actuator or sensor involved.

For a continuous statement, define a reference full-coordinate IVP by $M=M_0+M_r$, $K=(53/50)K_0$ and $D=\alpha_{\mathrm{shift}}M_0$, using the exact saved coefficients. It starts at the fresh seed's own saved numerical position and velocity and uses the same three executed fields and exact saved switching times. A piecewise cubic Hermite reconstruction and a residual estimate in the mechanical energy norm enclose this IVP without resetting the error at a switch. The reconstruction retains every control endpoint; each retained cell is bounded over its whole duration. Supplementary Section S17.2 gives the residual argument, and Section S17.3 specifies its numerical reception and the reconstruction choices, including unsuccessful bounds.

The preassigned port allowance remains $\varepsilon_y=10^{-8}\,\mathrm{J\,T^{-1}s^{-1}}$. At the first two hold starts, outward bounds on the discrepancy between the reference-IVP port and the actual logged readout are respectively at most 10^{-50} and less than $6.874\times10^{-9}\,\mathrm{J\,T^{-1}s^{-1}}$. Both nonzero proposals therefore satisfy the required observation premise. The bound at the third start is approximately 1.640×10^{-8} in the same units and exceeds the allowance. That hold applies exactly zero total field, so $\dot{H}_*=-v^TDv\le0$ and its energy argument needs no port observation. No three-observation accuracy claim is made.

For the same checkpoint, exact initial-ledger and command calculations give

$$\begin{aligned} L_0/E_{\mathrm{net}} &= 0.836050803, \\ L_1/E_{\mathrm{net}} &= 0.870553256, \\ L_2/E_{\mathrm{net}} = L_3/E_{\mathrm{net}} &= 0.910436431. \end{aligned} \tag{51}$$

Combining the two nonzero prefix-work bounds with the zero-field dissipation identity proves

$$H_*(t) \le 0.910436431 E_{\mathrm{net}} < 2.05 E_{\mathrm{net}}, \quad t_0 \le t \le t_3, \tag{52}$$

for this specified reference IVP. The proof uses the stated quadratic-family assumption, positive semidefinite damping, the same network's potential-value allowance, and the actual logged times and commands. It is not an enclosure theorem for COMSOL's compiled equations or dense output. Figure 2 displays the separate numerical execution. This finite execution and reference certificate supply no physical sensor guarantee, arbitrary future feedback guarantee, monotone energy claim for nonzero holds, or comparative control-performance claim.

8 Discussion

The received intervals give a precise interpretation to the fit of the selected complex model. At eleven ideal design configurations, the saved potential satisfies all 33 componentwise force constraints and all 31 coefficient constraints derived from them. Independent outward evaluation also bounds the actual coefficient loss without replacing it by zero. These results connect finite observations to one specified function. They do not require optimizer convergence, exact least-squares optimality or independent trajectory rows.

The observation precision limits what can be inferred about nonlinear constitutive behavior. The initial eleven-configuration point-force fit has a certified quartic discrepancy exceeding 6.3458×10^{-4} despite its small training score. Coefficient-guided training and final readout calibration substantially reduce that discrepancy, but the data-based quartic error allowance remains approximately 2.067×10^{-6}. The largest prescribed magnetic quartic is only approximately 5.364×10^{-9}, and every received total quartic interval contains zero. This sign ambiguity is an information diagnostic, not a calculation of the Birkhoff twist. Exact intersection of all force and coefficient inequalities also admits a scalar stiffness correction. Thus the finite data cannot require a nonlinear correction, even when a nonlinear network satisfies every received interval.

The inference gap now has both a coherent negative example and an explicit positive tolerance. A single quartic material correction respects the zero fifth-derivative condition and every original force and coefficient interval while cancelling the total fourth jet. A separate even polynomial is exactly invisible at all retained signed-node forces but has a nonzero target tail. The latter example shows why analyticity alone cannot supply the receiver's tail bound. For analytic perturbations with the quadratic jet fixed exactly, (35) gives a nonzero sufficient KAM neighborhood and includes the cubic contribution. The directly comparable observed material boxes do not receive that neighborhood. These statements delimit the current inference without replacing the unknown target by a declared family or by the selected network.

The candidate side of the regularity interface is also quantitative. Separate bounds on its third, fourth and fifth real derivatives have been received from the actual energy-feature graph on the full configuration cubes $|z|_\infty \le 1/2$ and $|z|_\infty \le 1/64$ (Supplementary Section S3). They provide the candidate terms of the regularity majorants in the force-to-jet argument. They provide no corresponding derivatives of an unknown material, population-risk bound, sampling-density hypothesis or uniform complex neighborhood for a learned class. Consequently the learning-theoretic and target KAM premises remain separate conditions.

The calibrated readout should be interpreted in light of that ambiguity. The final step changes the energy skip and 24 real coordinates of the last complex energy weight while preserving the trained hidden layers and dissipation branch. It performs one linear block solve and no gradient update. The readout norm decreases markedly, and the resulting function is compatible with a nearly scalar correction. This outcome preserves the 918-parameter function graph and its preceding full-layer training, but it provides no evidence that the complex representation is needed by these observations. The 31 coefficient rows are deterministic transforms of the same 33 force components, so their joint use in a weighted objective does not create 64 independent observations. All stages use development data; there is no independent validation or statistical generalization claim.

The deterministic loss bound also separates observation information from constitutive assumptions. The coefficient centers are obtained from the observed force labels, with their propagated errors and exported-center rounding retained. Vanishing cubic or quartic corrections are not added as training labels. The zero fifth-derivative remainder in the instantiated target bound follows from the explicitly stipulated quadratic material family. For an

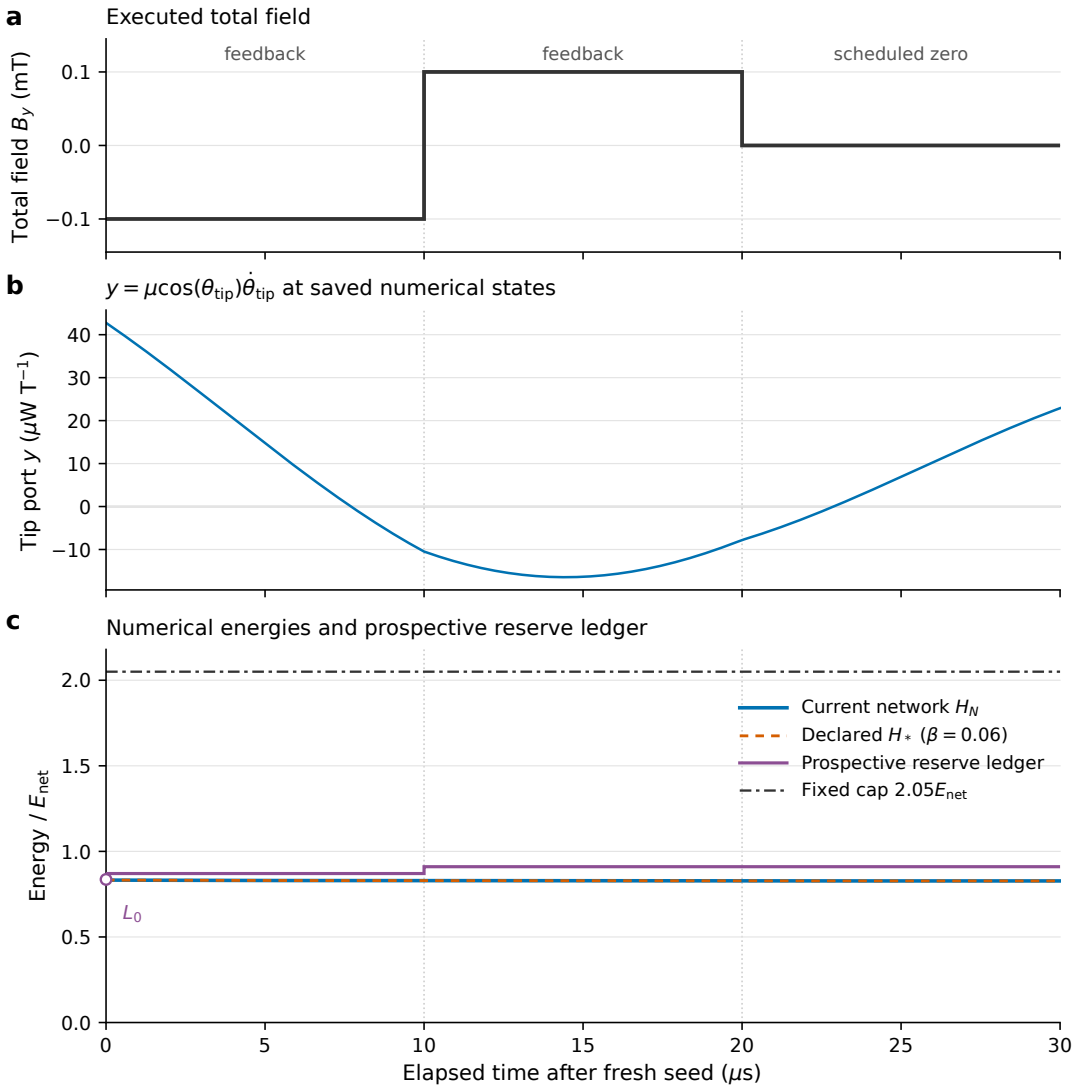


Figure 2. Executed total field, saved-state mechanical port, and numerical energies over the three approximately $10\,\mu$s holds, measured from the fresh numerical seed. The two nonzero fields are chosen from current observations; the third field is scheduled to zero. The learned and declared quadratic-family energies nearly overlap on this scale. The open marker denotes L_0; on each hold the reserve line displays $\max(L_k, L_{k+1})$. The curves connect saved numerical samples. The continuous cap in (52) is proved for the separately specified reference IVP by residual bounds and the work ledger, rather than inferred from these plotted samples.

unknown constitutive law, that remainder and any true-state measurement errors would have to be bounded independently. A finite coefficient bound alone supplies neither a uniform analytic error norm nor the target's nondegeneracy conditions.

The conservative certificate has a correspondingly specific scope. The same saved candidate has a positive equilibrium Hessian, the required finite-order frequency gaps and a nonsingular cubic-corrected quartic Birkhoff twist. Together with its verified analyticity, these conditions give positive-measure families of invariant three-tori for its three-coordinate restriction. A source-bound calculation additionally verifies these conditions for the full 64-coordinate neural Hamiltonian. It uses strict modal residual and projection bounds before enclosing the actual cubic-corrected twist, rather than borrowing the target family's quartic formula. The resulting 64-torus family has positive 128-dimensional measure in every neighborhood of equilibrium. Neither calculation gives a numerical torus radius. A separate calculation, conditional on $\mathcal{W}_* = \beta E_0$, verifies the local KAM conditions uniformly for every member of the admitted scalar interval (Supplementary Corollary S22). It uses the target family's analytic twist rather than the learned twist. This direct family theorem does not establish a perturbative transfer from the candidate or identify unknown physical constitutive behavior. An additional full-coordinate calculation establishes the same qualitative conclusion directly for the 64-coordinate transverse quadratic family, using exact finite spectral bounds, nonzero tip overlaps and the general-dimensional twist formula (31). This avoids assuming invariance of the three-coordinate trial space. The actual force-loss block is also connected to this family conclusion through

an explicit nonzero sufficient budget: $L_F \leq 12/275$ gives a larger parameter set whose full-transverse conditions are recomputed and pass. The direct data set is contained in that set; the learning-dependent sufficient test does not create identification information beyond its source data and constitutive prior. The full-coordinate candidate result is established independently of that loss-to-family interface. Neither argument includes axial modes or gives a numerical torus radius. None of these conservative results includes damping or time-dependent magnetic forcing.

The full-field safety statement uses different information. The candidate has an exact port-power identity, nonnegative dissipation and a separately checked sufficient coercivity condition. For a physical plant sharing the stated mass and full-state magnetic port, a uniform potential-value allowance transfers an energy guard under actual continuous saturated feedback. The instantiated quadratic-family guard uses $E_0 < 4E_{\mathrm{net}}$, cap $2.05E_{\mathrm{net}}$ and potential allowance $0.003838998456E_{\mathrm{net}}$. All 22 recorded numerical initial states satisfy this unchanged guard, with $0.237377387997 < H_N/E_{\mathrm{net}} < 0.972247611362$ and initial margin greater than $1.073913390182E_{\mathrm{net}}$ (Supplementary Corollary S33). This numerical reception supplies no error bound to a true physical initial state. Family membership is a conditional physical assumption, not a parameter identification conclusion. Likewise, the candidate's dissipation lower bound on its feature box is not a bound on unknown physical damping or on a different configuration domain.

The sampled-work theorem supplies a sufficient condition for held inputs. With the additional physical bound $D_* = a(x)D_0$, $0 \leq a \leq 6/5$, it bounds intersample port drift and charges held-input work to a strict energy ledger. At the fixed 10^{-4} T total-field limit, all 22 initial nonzero proposals pass for both 1 and 10 μs holds; none passes for 100 μs. The latter rejection is a decision of this sufficient guard, not evidence of unsafe motion at that duration. A separate native three-hold feedback execution is completed, and the continuous reference-IVP cap in (52) receives the two nonzero hold-start errors and the zero-input dissipation branch. The third start-port error bound exceeds the fixed allowance and is not used by that zero branch. The arbitrary-horizon energy conclusion requires the stated sensing, timing and actual-command conditions at every subsequent hold; a finite set of accepted initial proposals does not verify those conditions. In particular, the $10^{-8}\,\mathrm{J\,T^{-1}s^{-1}}$ port allowance is received for the reference IVP at the two executed nonzero hold starts. It remains unverified for a physical sensor. The residual reconstruction encloses the declared equation with its own saved seed, rather than COMSOL's compiled flow or dense output.

The distinction between full-field observations and the conservative restriction remains essential throughout. A projected spatial trajectory can retain unresolved elastic and dissipative forces, so the three-coordinate restriction is not asserted to close every full-field motion autonomously. The force-label reception controls the particular observation interface used here. It does not establish a general trajectory approximation theorem, continuum discretization accuracy or equivalence between a stored reconstruction and all operations of a compiled simulator. The certificates interpret the saved binary weights and mechanical coefficients as exact constants in their specified graphs; rounded runtime evaluation is a separate numerical object.

Extending these results to an unknown device therefore requires both more informative observations and an implementation contract. A useful configuration and velocity design should resolve the coefficient signal after propagating observation errors, while independent records and constitutive remainder bounds should test behavior beyond the fitting design. Physical use of the safety theorem additionally requires sound state and port estimates and control of unmodelled work. The present sampled theorem supplies an intersample work and ledger analysis under explicit conditions; physical use must establish those conditions, including observation errors at the actual hold starts and execution of the accepted total field. Delays or actuation errors outside this contract need additional bounds. These are concrete conditions for transferring the present model-level results to a robotic experiment.

Data and code availability

The accompanying reproducibility manifest identifies the native simulation records, selected network parameters, exact operator records and interval verification sources used for each numerical statement. It separates local source files from workspace evidence and records their SHA-256 identities. The supplementary material supplies the mathematical proofs and arithmetic conventions. Simulation data are not represented as physical measurements, and no independent test dataset is claimed.

Declarations

This study reports mathematical analysis and simulated mechanical data. It involved no human participants or animals.

References

Abbott JJ, Diller E and Petruska AJ (2020) Magnetic methods in robotics. *Annual Review of Control, Robotics, and Autonomous Systems* 3: 57–90. doi: 10.1146/annurev-control-081219-082713

Chen Y, Matsubara T and Yaguchi T (2022) KAM theory meets statistical learning theory: Hamiltonian neural networks with non-zero training loss. *Proceedings of the AAAI Conference on Artificial Intelligence* 36(6): 6322–6332. doi: 10.1609/aaai.v36i6.20582

Desai SA, Mattheakis M, Sondak D, Protopapas P and Roberts SJ (2021) Port-Hamiltonian neural networks for learning explicit time-dependent dynamical systems. *Physical Review E* 104: 034312. doi: 10.1103/PhysRevE.104.034312

Greydanus S, Dzamba M and Yosinski J (2019) Hamiltonian neural networks. In: *Advances in Neural Information Processing Systems*, vol. 32. https://arxiv.org/abs/1906.01563.

Haro À and Luque A (2019) A-posteriori KAM theory with optimal estimates for partially integrable systems. *Journal of Differential Equations* 266(2–3): 1605–1674. doi: 10.1016/j.jde.2018.08.003 Preprint version used: https://arxiv.org/abs/1711.05515v1.

Kim Y, Parada GA, Liu S and Zhao X (2019) Ferromagnetic soft continuum robots. *Science Robotics* 4(33): eaax7329. doi: 10.1126/scirobotics.aax7329

Pöschel J (2001) A lecture on the classical KAM theorem. In: *Proceedings of Symposia in Pure Mathematics*, vol. 69, pp. 707–732. doi: 10.1090/pspum/069/1858551 Preprint version used: https://arxiv.org/abs/0908.2234v1 (Version 1.3, 2009).

Trabelsi C, Bilaniuk O, Zhang Y et al. (2018) Deep complex networks. In: *International Conference on Learning Representations*. https://arxiv.org/abs/1705.09792.

Zhang SQ, Gao W and Zhou ZH (2022) Towards understanding theoretical advantages of complex-reaction networks. *Neural Networks* 151: 80–93. doi: 10.1016/j.neunet.2022.03.024

Zhong YD, Dey B and Chakraborty A (2020) Dissipative SymODEN: Encoding Hamiltonian dynamics with dissipation and control into deep learning. In: *ICLR Workshop on Integration of Deep Neural Models and Differential Equations*. https://arxiv.org/abs/2002.08860.